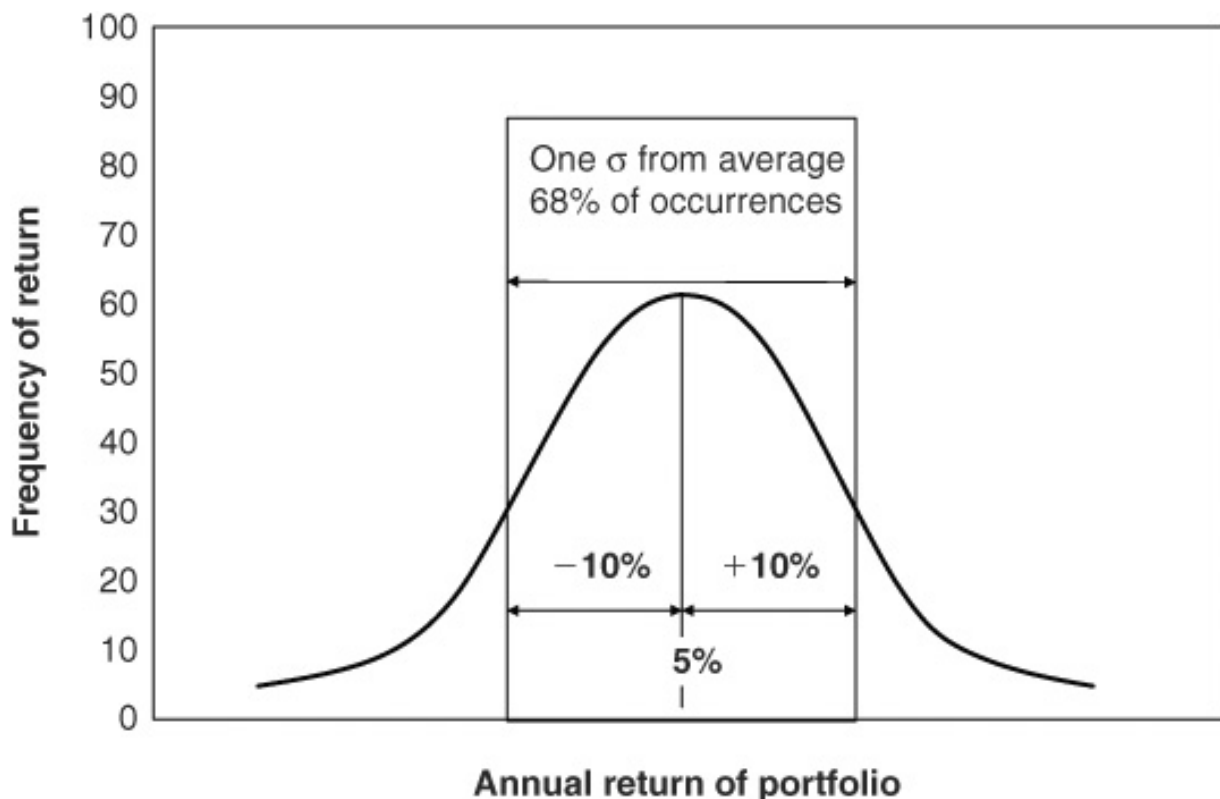


often people look at their account value in a bear market, the more apt they are to do something that is not in their long-term best interest.

Volatility is measured in units of standard deviation; the larger the variation in value, the higher the standard deviation. Standard deviation (expressed as the Greek symbol σ) measures the average amount of data discrepancy around the data average. For example, assume that an investment has an average annual return of 5 percent and an annual standard deviation of 10 percent. This means that the average annual deviation from the 5 percent is 10 percent. Sometimes it is more and sometimes less, but on average it is 10 percent. In a “normal distribution” of annual returns over time, there is a 68 percent chance that an annual return will fall between -5 percent and +15 percent. [Figure 2-3](#) illustrates this concept.

FIGURE 2-3
Normal Distribution



The standard deviation of asset-class returns is not static. There are times when the returns are more volatile than others. [Figure 2-4](#) shows the rolling five-year standard deviation of returns for various asset classes.